Souvick Bera

PhD Candidate , Applied Mathematics and Statistics
Colorado School of Mines

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Research Interests

Time Series, Spatial Statistics, Spectral Analysis, Resampling Methods, Empirical Likelihood, Climate Uncertainty.

Education

2022–2027 **Ph.D., Statistics**
Colorado School of Mines, Golden, CO, USA
Advisor: Dr. Soutir Bandyopadhyay and Dr. Haihan Yu

2018–2020 **M.Sc, Statistics**
University of Calcutta, Kolkata, India

2015–2018 **B.Sc, Statistics**
RKMRC Narendrapur, Kolkata, India

Professional Experience

2021–2022 **Research Assistant**
IIMA, Ahmedabad, India
Supervisor: Prof. Hyokjin Kwak and Prof. Tathagata Bandyopadhyay

- Worked on applied marketing analytics research using statistical methods to extract insights from real world marketing data. This work involved data preparation and cleaning, building and validating econometric and probabilistic models, and aligning empirical findings with marketing literature to support data-driven decision making.

Publications

Peer-Reviewed Journal Articles

1. **Souvick Bera**, Daniel J. Nordman, and Soutir Bandyopadhyay. (2026). Frequency Domain Resampling for Gridded Spatial Data. *Statistica Sinica*, (Accepted). [DOI] [arXiv]

In Preparation

1. **Souvick Bera**, Haihan Yu, Daniel J. Nordman, and Soutir Bandyopadhyay (20xx). Demystifying Spectral Bias in Irregularly Spaced Spatial Data. *To be Submitted* to *The Annals of Statistics*. (Draft available upon request).
2. **Souvick Bera**, Debdeep Pati, and Soutir Bandyopadhyay (20xx). Distributionally Constrained Frequency-Domain Empirical Likelihood. *In Preparation* for *Journal of the American Statistical Association*.
3. **Souvick Bera**, Nathan Lenssen, and Kevin Schwarzwald (20xx). Partitioning Variability in

Climate Projections from Unbalanced Ensembles. *In Preparation* for *Geophysical Research Letters*. (Draft available upon request).

Honors and Awards

Oct 2024	ENVR Travel Support, ASA ENVR.
May 2025	Departmental Travel Award, Applied Mathematics and Statistics, Colorado School of Mines.
Spring 2025	Graduate Student Teaching Award, Department of Applied Mathematics and Statistics, Colorado School of Mines.
May 2026	Maurice Davies Award, ASA CO-WY Chapter.
July 2026	GSG Travel Award, Colorado School of Mines.
Aug 2026	Departmental Travel Award, Applied Mathematics and Statistics, Colorado School of Mines.
Sep 2026	ENVR Travel Support, ASA ENVR.

Presentations

Contributed/Invited Talks and Posters

July 2023	Variance Estimation of Spatial Spectral Statistics using Subsampling, Spatial Statistics: Climate and the Environment, Boulder, CO. *(Poster).*
Aug 2024	Frequency Domain Resampling for Spatial Data, Joint Statistical Meetings, Portland, OR. *(Contributed Talk).*
Oct 2024	Robust Inference Using Empirical Likelihood, ENVR Workshop: Spatial Data Science for the Environment, Boulder, CO. *(Poster).*
June 2025	Frequency Domain Resampling for Gridded Spatial Data, IISA Conference, Lincoln, NE. *(Contributed Talk).*
May 2026	Frequency Domain Resampling for Spatial Data, New England Science Symposium, Storrs, CT. *(Invited Talk).*
Aug 2026	Distributionally Constrained Frequency-Domain Empirical Likelihood, Joint Statistical Meetings, Boston, MA. *(Contributed Talk).*
Oct 2026	Distributionally Constrained Frequency Domain Empirical Likelihood, ENVR Workshop, Houston, TX. *(Poster).*

Teaching

Course Instructor

2023–2025	**MATH 201: Introduction to Statistics**, Colorado School of Mines. *Fall 2023, Spring 2024, Fall 2024, Spring 2025.*
Spring 2026	**MATH 324: Statistical Modeling**, Colorado School of Mines.

Teaching Assistant

2022–2023	**MATH 225: Differential Equations**, Colorado School of Mines. *Fall 2022, Spring 2023.*
2024–2025	**MATH 530: Introduction to Statistical Methods**, Colorado School of Mines. *Summer 2024, Summer 2025.*

Professional Service

Student Volunteer

2023	Student Volunteer, *International Indian Statistical Association (IISA) Conference*, Golden, CO.
2024	Student Volunteer, *Extremes* 2024, Golden, CO.
2026	Student Volunteer, *ASA CO-WY Chapter*, Golden, CO.

Technical Skills

Programming	C++, R, Python, MATLAB.
Statistical Software	R (`tidyverse`, `ggplot2`, `parallel`, `doParallel`), Python (`NumPy`, `multiprocessing`, `SciPy`, `scikit-learn`)
High-Performance Computing	Parallel computing and large-scale simulation studies on HPC clusters using Slurm.
Tools	Git/GitHub, Linux, VS Code, Jupyter, RStudio, Overleaf
Languages	Bengali (native), English (fluent), Hindi (fluent)

Professional Memberships

American Statistical Association (ASA), Institute of Mathematical Statistics (IMS), Mathematical Association of America (MAA).

References

Dr. Soutir Bandyopadhyay (Ph.D. Advisor)
Professor
Applied Mathematics and Statistics
Colorado School of Mines
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Dr. Haihan Yu (Ph.D. Advisor)
Assistant Professor
Computer Science and Statistics
University of Rhode Island
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Dr. Daniel J. Nordman (Collaborator and Ph.D. committee member)
Professor
Department of Statistics
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Dr. Nathan Lenssen (Collaborator)
Teaching Assistant Professor
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Last updated: October, 2026